LEF MODEL · CHAPTER 22

Chapter 22: A Full Planetary Survey — from Mercury and Venus to Pluto, Haumea and Ceres

Based on Paper 11 — 11. Moon Births (Parts C–N: applying the protocol of 13 checks to all the planets and five dwarf planets, the capture rule and the 12 open questions)

01 · THE MYSTERY

The mystery: how do 13 bodies in the solar system stand up to the test of the Moon Law?

In Chapter 21 we formulated **the Moon Law**, the principle of the hydrodynamic pump and the principle of the atmospheric lid. In Parts C–N of Paper 11 the strict protocol of 13 checks is run on **all eight planets and five dwarf planets** to test whether there is a single body that contradicts the mechanism:

1. **The hot inner edge (Mercury, Venus, Earth, Mars — Parts C–F):** why were Mercury and Venus left without moons, and how was their corotation line pushed *outside the Hill sphere*?
2. **The gas and ice giants (Jupiter, Saturn, Uranus and Neptune — Parts G–J):** why at Jupiter does the density fall with distance (from Io to Callisto) while at Uranus it rises (from Miranda to Oberon)? How do Saturn's rings look young although they are fed by an ancient process? And what really tilted Uranus by 98°?
3. **The cold outer edge (Pluto–Charon, Haumea, Eris, Makemake and Ceres — Part K):** how did inner differentiation tear apart the Pluto–Charon peanut,

why does Haumea stand just before a split, and why does Ceres have no moons?

02 · AN IMAGE

An image: Saturn's rings as a fountain in equilibrium, a captured peanut lobe, and a spinning top whose axis tips over

Four tangible images explain the results of the check of the planets and the dwarf planets:

- **A fountain pool in which the filling rate equals the draining rate (Saturn's rings as a continuous flow, Part H):** Cassini measured an enormous rate of "ring rain" (hundreds to thousands of kg per second) falling from the rings onto Saturn — a rate that would have destroyed the rings within a few hundred million years had they been static. In the LEF model, Saturn's gas lid (70%–85%) is not completely sealed: the active inner pump leaks water to the corotation line (1.92 R, which runs exactly inside ring B!) at a rate equal to the rate of the falling rain — so the ring looks young and clean although its engine is ancient!
- **Two torn peanut lobes — one captured by Neptune and one thrown into space (Triton, Part J):** Triton orbits Neptune in a retrograde orbit (157°) and its density is high (2,061 kg/m³, heavier and denser than Pluto). In the LEF model, Triton is **the large lobe of a peanut system that differentiated and was torn apart in the Kuiper belt** (which is why it kept inside it the compact silicate-metal core), was captured by Neptune's gravity and destroyed its original system of regular moons!
- **A spinning top that widens the edge of its skirt and enters resonance with the Sun (Uranus' 98° tilt, §I.1):** a vector calculation proves that the escape of water from the equator takes away angular momentum *parallel to the rotation axis only* (it shortens the arrow from 8.3 h to 17.2 h, but does not turn it). By contrast, a mass of 2% of Uranus' mass in the form of water and moons floating in the equatorial plane at a distance of 20 R enlarges the effective flattening moment $J_{2,\,\mathrm{effective}}$ **1,200 times**, brings the precession rate into

secular resonance with the orbital plane, and gradually tilts Uranus' axis to 98°!

- **A frozen ice shell that cracks all at once over a soft core (Charon's tectonic canyon, §K.1):** in the Pluto–Charon system, the unified body spun in 2.51 hours (close to the fission threshold of 2.44 hours). The sinking of the core reduced the moment-of-inertia coefficient k from 0.40 to 0.31 and sped the rotation up to 1.94 hours (x = 0.86): the frozen ice shell broke all at once along the waist, exposing the smooth plain of Vulcan Planitia, which always faces Pluto!

03 · SCIENCE AND MODEL

Science and model: the findings of the protocol of 13 checks from planet to planet (Parts C–K)

Below is the essence of the systematic check of all the systems in the solar system:

1. **Mercury and Venus (Parts C–D):** nearness to the Sun shrank the Hill sphere (220,600 km at Mercury, 1,010,600 km at Venus), and solar tidal braking slowed the rotation (58.6 days at Mercury, 243 days retrograde at Venus) until the corotation line left **the Hill sphere** (242,900 km at Mercury, 1,536,500 km at Venus). In addition, fast thermal drying (D/H > 100× at Venus) closed the birth window early.

2. **Jupiter (Part G):** the corotation line stands today at 160,000 km (2.29 R, identical to the Earth at birth!), and its diluted core (Juno) is still differentiating; but the enormous gas lid (90%–97%) condenses the water at 5–7 bar and prevents new births. The four Galilean moons record the inward movement of the corotation line: Callisto was born first (1,834 kg/m³, undifferentiated, k = 0.355) and Io was born last (3,528 kg/m³, dry and metallic), while eight inner moons orbit on both sides of the line (1.83 R–3.17 R).

3. **Saturn (Part H):** the corotation line (1.92 R) crosses ring B; the spongy ring moons (400–640 kg/m³) lie just beyond the line (2.29 R–2.60 R); and Iapetus

(1,088 kg/m³) keeps a frozen bulge matching a rotation of 16 hours — that is, it coalesced and froze exactly at a distance of **2.53 R**, above the ancient corotation line, before it was pushed out to 61 R!

4. **Uranus and Neptune (Parts I–J):** the thin gas lid (10%–20%) allowed the escape of water amounting to at least 2%–3% of the planet's mass (and up to ~ 11%), which slowed Uranus from 8.3 to 17.2 hours and Neptune from 7.3 to 16.1 hours and pushed the corotation line from ~ 2 R to 3.24 R and 3.39 R. In doing so the line crossed the orbits of the inner moons (1.95 R–2.99 R at Uranus, 1.96 R–2.99 R at Neptune), which today are falling back inward! At Neptune, the moon Proteus (420 km, right on the threshold of rounding) stayed irregular because of high sponginess (prediction: an especially low density).

5. **The dwarf planets (Part K):** Pluto and Charon are a peanut torn apart at 1.94 hours (x = 0.86), and the four small moons (Styx, Nix, Kerberos and Hydra) were thrown from the peanut's equator before the tearing and pushed outward first by the peanut's quadrupole moment and later by the tide; Haumea (3.9 hours, 2,300 × 1,100 km, corotation line at 1,104 km inside the ends of the long axis at 1,161 km) stands just before a split; and Eris–Dysnomia is a scaled-down replica of Pluto–Charon.

`The objective scoring rule for identifying a captured moon (L.3)`

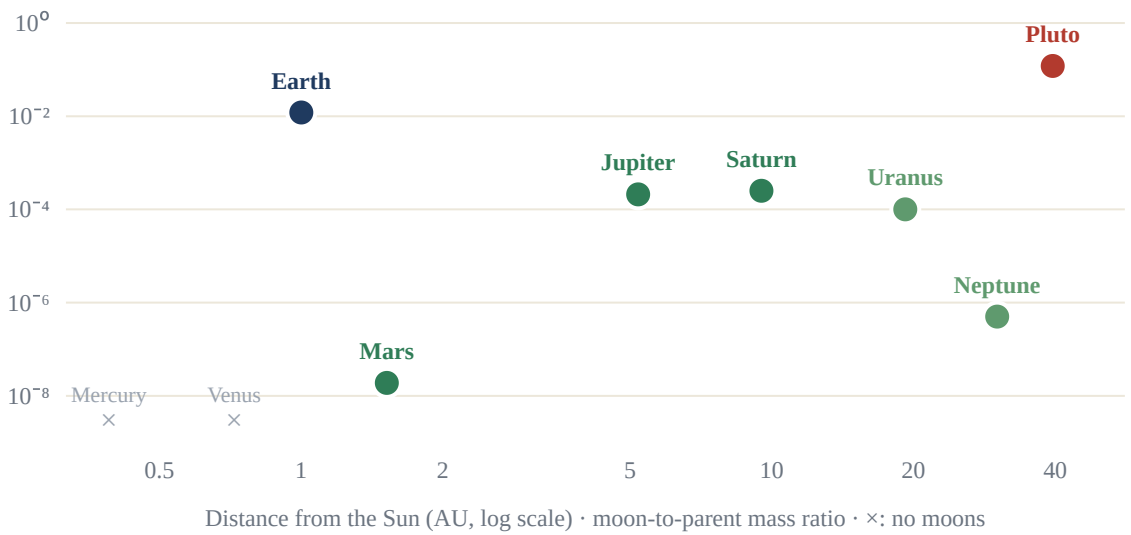
So as not to classify every anomalous moon as captured arbitrarily, a threshold of **≥ 3 points** is set: a retrograde orbit (>90° from the Laplace plane) = 2 pts, high inclination = 1, high eccentricity = 1, extreme distance = 1, and a different chemical composition = 1. Triton (3 pts), Phoebe (4 pts) and Sycorax (5 pts) are classified as captured; while Iapetus (0 pts) and the Earth's Moon (0 pts relative to the Laplace plane!) are original moons born from their parents.

The mechanism in words: the arc of births across the solar system (L.4) and the 12 open questions (Part M)

Parts L.4, M and N of Paper 11 sum up the global picture of the solar system and mark the open directions for future research:

1. **The arc of births as a function of distance from the Sun (L.4):** when the mass of the regular moons relative to the parent is mapped as a function of distance from the Sun, **a continuous arc** with three stations emerges:
    - 2 **The inner thermal edge (0.39–1.00 AU):** Mercury (0) and Venus (0) dried out and closed early, and the Earth (1.2×10^{-2}) barely made it and built one dry moon on the tip of a pear.
    - 3 **The zone of peak pumping (1.52–9.50 AU):** from Mars (2 moons, 1.9×10^{-8}), through Jupiter (8 moons, 2.1×10^{-4}), up to the peak at Saturn (~ 24 regular moons, 2.5×10^{-4}).
    - 4 **The frozen outer edge (19.2–39.5 AU):** at Uranus (19 moons, 1.0×10^{-4}) and Neptune (7 moons, ~ 5×10^{-7}, a 200-fold drop) the freezing of the crust gradually blocks the pump, and at Pluto (39.5 AU) the system passes to a full peanut split (5 moons, mass ratio 0.12).
5. **The link between the corotation line leaving the Hill sphere and the loss of the planet's own rotation (L.1):** Mercury and Venus are the only planets whose corotation line went outside the Hill sphere (100 R against 90 R at Mercury; 254 R against 167 R at Venus — exactly when $\Omega < \sqrt{3}\ n$, that is, the day is longer than 58% of the year!), and it is exactly in them that the Sun took over the rotation (3:2 locking at Mercury and retrograde rotation at Venus).
6. **The twelve open questions of Part M:** the paper gathers 12 open problems for calculation and measurement, among them: the reversal of the density order at Uranus (Miranda to Oberon); calculating the resonance windows for Uranus' 98° tilt; the location of Pluto's tearing scar (is the Sputnik Planitia

basin the tearing site?); separating the one-sided thickening component from the symmetric elongation component in the shape of the Earth's Moon; the gas-to-water ratio in the mass that escaped from Uranus and Neptune (1.6%–10.8%); pulsing pump activity at Saturn; the blocking of birth at Ceres (x = 2.54); active ejection today from the ends of Haumea; the formation of the equatorial ridges of Iapetus, Pan and Atlas as spatula ridges; and the almost-exact locking of Venus to the Earth (243.02 against 243.16 days).



The arc of births across the solar system (L.4): one large moon at the Earth, the ejection zone from Mars to Neptune, and a peanut split at Pluto.

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The effective flattening moment $J_{2,\,\text{effective}}$ governing the precession rate of the rotation axis (as at Uranus and Saturn) and the condition for the corotation line to escape outside the Hill sphere are formulated as follows (§I.1, §L.1):

$$J_{2,\,\text{effective}} = J_2 + \frac{1}{2}\sum_i \frac{m_i}{M}\left(\frac{a_i}{R}\right)^2$$

$$a_c > r_H \iff \Omega < \sqrt{3}\; n$$

When water or moons stay temporarily in an equatorial orbit at a distance of $a_i \approx 20R$, the term $\frac{1}{2}(m_i/M)(a_i/R)^2$ enlarges Uranus' effective flattening moment 1,200 times (from 0.0033 to about 4.0) and brings the precession rate into a secular resonance that tilts the axis to 98°; while when mass escape slows an inner planet down to $\Omega < \sqrt{3}\ n$, the corotation line leaves the Hill sphere and the Sun takes over its rotation.

06 · THE SIGNATURE

The signature in reality: reconstructing the corotation line and the mass that escaped from each planet (Table L.1)

A full calculation of angular-momentum conservation (including rotational flattening by the Darwin–Radau method and the shrinking of the giants, script `p12_line_then.py`) divides the solar system into three clear groups (§L.1):

- **The gas giants with a thick lid (Jupiter and Saturn):** conservation of angular momentum alone places the corotation line at the time of birth at 2.35–2.57 R at Jupiter and 1.96–2.42 R at Saturn — almost no shift compared with today (2.29 R and 1.92 R), and a mass escape of only 0%–1.0%.
- **The ice giants and Mars (a thin lid or exposed land):** conservation of momentum without mass loss leaves the line far from the birth range (7.33 R at Mars, 3.24–4.33 R at Uranus, 3.39–4.40 R at Neptune) — so an escape of water and gas through the corotation line is required, amounting to **6.8%–16.1% of the mass of Mars**, **1.6%–10.7% of the mass of Uranus**, and **1.8%–10.8% of the mass of Neptune**!

The lab box: the corotation line then and now, the mass that escaped and the arc of births (Paper 11, Tables L.1 and L.4)

Planet	Corotation line today (R)	Corotation line then without mass loss	Corotation line at the time of the births (R)	Day at the time of the births (h)	Mass that left through the line (% of the planet's mass)
Mercury / Venus	100 / 254 (outside Hill!)	Cannot be reconstructed	No birth window opened	—	1.5%–9.6% (gas + solar braking)
Earth	**6.62**	**2.54–2.59**	**2.33 (2.29)**	**5.77**	**1.0%–1.8% (ancient gas envelope)**
Mars	**6.03**	**7.33**	**1.89–2.16**	**7–9**	**6.8%–16.1% (water and steam)**
Jupiter	2.29	2.35–2.57	**2.35–2.57**	11.8–19.5	**0%–1.0%** (thick gas lid)
Saturn	1.92	1.96–2.42	**1.96–2.42**	12.6–24.7	**0%–0.5%** (thick gas lid)
Uranus	**3.26**	**3.24–4.33**	**1.75–2.30**	**6.3–14.9**	**1.6%–10.7% (thin lid, water escape)**
Neptune	**3.39**	**3.39–4.40**	**1.75–2.30**	**5.5–13.1**	**1.8%–10.8% (thin lid, water escape)**

Parent	Distance from the Sun (AU)	Birth mode	Number of regular moons	Ratio of moon mass to parent ($m_{\text{moons}}/M_{\text{parent}}$)
Mercury / Venus	0.39 / 0.72	No birth (fast thermal drying)	0	0
Earth	**1.00**	**Built on a pear tip**	**1**	**1.2×10^{-2} (the peak among the planets)**
Mars	1.52	**Ejected on a bulge**	2	1.9×10^{-8}
Jupiter	5.20	**Ejected**	8	2.1×10^{-4}

Saturn	**9.50**	**Ejected — the peak of the pump**	**~ 24**	**2.5×10^{-4} (the peak of ejection)**
Uranus / Neptune	19.20 / 30.10	**Ejected — crustal freezing**	19 / 7	1.0×10^{-4} / ~ 5×10^{-7}
Pluto	**39.50**	**Full peanut split**	**5**	**0.12 (the peak of the solar system)**

07 · NEXT

The door ahead

In Chapters 21–22 we applied **the Moon Law** and the protocol of 13 checks to the whole solar system — from moonless Mercury and Venus, through the Earth and Mars, the gas and ice giants (Jupiter, Saturn, Uranus and Neptune), to Pluto, Haumea and Ceres — and showed how the pump principle and the atmospheric-lid principle unite them all under one law.

Chapter 23

The Planetary Rings — the Source of the Material, the Threshold Day of 13.4 Hours and Saturn's Equilibrium Flow →

← Chapter 21 · Prologue

© 2026 Zohar Duran · The LEF Model · lef-model.com · ORCID 0009-0005-3194-9387